

Weekly Progress Report

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Domain: **Data Science and Machine Learning**

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Week Ending: 04

I. Overview:

This week, the primary focus was on progressing the project Quality Prediction in a Mining Process from a simple functional prototype to an industry-level prototype application. The development concentrated on improving the application's functionality, reliability, user interaction, data processing, validation, and presentation of quality-related results. The objective was to transform the initial prototype into a more structured and practical application suitable for real-world industrial use.

II. Achievements:

1. Industry Level Application Development

Enhanced the initial prototype into a structured industry-level application with an improved user interface and organized workflow. Integrated appropriate input fields and controls for important mining quality parameters and developed a more systematic process for entering, processing, and evaluating sample data.

2. Quality Assessment and Processing

Strengthened the application's core processing logic to evaluate important quality parameters and generate meaningful quality results. Improved the handling of user inputs and incorporated validation checks to reduce incorrect or unrealistic data entry.

3. Data Visualization & Application Enhancements

Improved the application's data presentation by incorporating interactive visualizations and clearly organized quality metrics. Enhanced the overall

application structure to make important information easier to interpret and support faster decision-making.

4. Validation and Reliability

Introduced validation and evidence-related components to improve the reliability of the application's results. Added mechanisms for comparing calculated quality classifications with reference information and maintaining supporting information for the assessment.

5. Economic Feasibility Integration

Extended the prototype beyond basic quality prediction by incorporating economic feasibility-related parameters. The application can process relevant cost, selling price, margin, and economic information to provide a broader assessment of the mineral sample.

III. Challenges:

1. Converting The Prototype into an Industry-Level Application

The major challenge was transforming the initial simple prototype into a more structured and reliable application. This required improving the application's workflow, input validation, calculations, result presentation, and overall usability.

2. Data Validation and Reliability

Ensuring that user-provided sample data was within acceptable ranges and that the calculated results were reliable required additional validation logic. Maintaining appropriate reference information and supporting evidence was also an important consideration.

3. Integration of Multiple Functionalities

Integrating quality assessment, validation, visualization, reporting, and economic feasibility into a single application required careful organization of the application's processing workflow and data structure.

IV. Learning Resources:

1. Application Development and UI Design

Studied Streamlit documentation and application development resources to improve the structure, layout, interactive components, and performance of the application.

2. Data Processing and Validation

Explored techniques for structured data processing, input validation, database management, and reliable calculation workflows for developing an application suitable for practical industrial applications.

3. Reporting and Data Management

Learned methods for integrating structured reports, data storage, and export functionality to make the application more useful for maintaining and communicating assessment results.

V. Next Week's Goals:

1. Industry-Level Application Refinement

Further improve the application by refining its interface, workflow, validation mechanisms, and overall usability. Focus on making the application more robust and suitable for practical industrial scenarios.

2. Testing and Validation

Perform comprehensive testing using different sample inputs and conditions. Verify the quality assessment, compliance results, validation process, and economic calculations to identify and correct potential errors.

3. Documentation and Deployment

Prepare complete technical documentation describing the application's architecture, functionality, calculation methodology, validation approach, and usage workflow. Continue improving the deployment readiness of the application.

VI. Additional Comments:

- The project has progressed from a simple working prototype to an industry-level prototype application, with multiple integrated functionalities for quality assessment, validation, visualization, reporting, and economic feasibility.
- The current development phase focuses on improving reliability, usability, validation, and practical applicability rather than only demonstrating the basic concept.